

Unit 2 Quiz

Knowledge

Part A: Multiple Choice .Write the CAPITAL LETTER corresponding to the correct answer on the line provided.

1. If $f(x) = \sqrt{x^2 - 1}$ and, $g(x) = x + 1$ which expression is equal to $[g(f(x))]'$? _____
(A) $1 + \sqrt{x^2 - 1}$ (B) $\frac{x+1}{\sqrt{x^2 + 2x}}$ (C) $\frac{x}{\sqrt{x^2 - 1}}$ (D) $\sqrt{x^2 + 2x}$
2. For what value(s) of x do the tangents to the curves $f(x) = (x^2 + 1)^3$ and $g(x) = 3x^2$ have the same slope? _____
(A) $x = -\sqrt{2}, \sqrt{2}$ (B) $x = -1, 0, 1$ (C) $x = 0, \pm\sqrt{2}$ (D) $x = 0$
3. 4. The slope of tangent line to the curve $f(x) = \frac{4}{\sqrt[3]{x}}$ at point $x=8$ is: _____
(A) $-\frac{1}{48}$ (B) $\frac{1}{48}$ (C) $-\frac{1}{12}$ (D) $\frac{1}{12}$
4. Find $f'(-1)$ if $f(x) = ax + \frac{b}{x}$, _____
(A) $a + b$ (B) $a - b$ (C) $a + 2b$ (D) $-a + b$

Part B: Full Solution

5. Differentiate the following. Leave answers in simplified form.

a) $f(x) = (1 + \sqrt[4]{x^3})(1 + x^2)^2$

b) $f(x) = \frac{2x^2 + 6\sqrt[5]{x^2} + \pi x}{\sqrt[3]{x}}$

c) $y = \frac{(3x^2 - 5)^3}{\sqrt[3]{x+1}}$

d) $y = x + \sqrt{x^2 + \sqrt{x^2 - x}}$

6. Given $y = u^3 + \frac{1}{2}u^2$ and $\frac{dy}{dx} = -20$ when $u = 1$, find the value of $\frac{du}{dx}$.

Application

1. If $f(x) = \sqrt[3]{x^2} + \frac{4}{(x-2)^2}$, find $f''(1)$.
2. Given $f'(1) = 4$, $g'(1) = -2$, $f(1) = 1$, and $g(1) = 1$ find $h'(1)$ if $h(x) = xg(f(x^2))$.
3. Consider the graph of $f(x) = x^3 - 9x$. The tangent has a y-intercept of 16. Determine the exact area of the triangle formed in the second quadrant by the tangent line and the axes.

4. A particle moves according to the equation $s(t) = 4\left(\frac{4}{3} - \frac{1}{35}t^2\right)t^{\frac{3}{2}}$ where t is in seconds and $s(t)$ is the distance in metres. Determine the following:
- a) the velocity of a particle when its acceleration is zero
- b) exact time(s) at which the velocity is equal to zero.

Thinking

1. Consider the rational function $f(x) = \frac{ax-b}{bx+1}$, where a and b are constants. Determine values for **a** and **b** such that the tangent lines to the graph of $f(x)$ at points $x=0$ and $x=-\frac{2}{3}$ are parallel to the line $y=5x-2$.